Projector-Resolved Seed Calculus II

Polyphase Refinement, Weighted Transfer, and Dyadic–Odd Resolution

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Abstract

This paper develops the refinement layer of the projector-resolved seed calculus. A packet decomposition is promoted from one coordinate choice to a coherent tower of exact coordinate systems: coordinate changes satisfy identity and cocycle laws, and typed operator matrices are natural under simultaneous source and target refinement. Coarse packets are sums of fine descendants, operator blocks aggregate exactly across levels, and residue selectors admit both division-free coefficient formulas and finite-Fourier realizations over a commutative semiring from a finite orthogonality table. For $N = 2^a M$ with M odd, the Chinese remainder theorem separates dyadic and odd packet addresses. Weighted transfer inversion is proved in a certified finite quotient and in a complete normed ring under a strict contraction; the finite theorem constructs the signed, cumulative-weight-pruned word sum and permits a sharper length cutoff from a uniform lower edge weight. For a finite positive-weight multigraph, Lean proves the first-edge recurrence for the enumerated walk series and identifies it with the automatically generated determinant/adjugate expression, yielding rationality and a recurrence without a supplied rationality certificate. Minimal evaluated Bézout hypotheses give a quasi-idempotent numerator and an idempotent separator, while explicit lattice-unit data gives an upward-closed separation-weight bound. Concrete $\mathbb{Z}/p^L\mathbb{Z}$ Smith fibres have the expected divisibility criterion and exact cardinality; the existence of Smith normal form itself remains classical input. The rational sign split has exact 2-local denominator weight one, equal to the 2-adic valuation of its computed resultant, and exact local weight zero at odd primes. Typed linear defects propagate through powers, polynomial actions, and explicitly witnessed resolvents, with proved operator-norm majorants. The companion release contains thirteen generated figures, a versioned Python implementation with forty-eight tests, and a Lean 4/Mathlib formalization of the frozen twenty-five-item contract. These checks establish the stated formal contract, not novelty, priority, or external peer review.

Scope. Paper I [1] fixed the typed seed, operator, relation, packet, completion, and reconstruction language. Paper II assumes that foundation and studies exact refinement, coordinate-change coherence, coefficient and phase selectors, certified weighted inverses, finite weighted-walk enumeration and rationality, evaluated Bézout separators, Smith-coordinate fibres, and typed linear defects. The transfer-zeta construction is automatic only for the finite, positive-integral-weight graph model stated here. Universal complete-filtered-algebra convergence, construction of a general polynomial resultant certificate, and the structure theorem producing Smith normal form remain cited or explicit input rather than being hidden inside stronger claims. The concrete sign-factor resultant is computed separately. Nonlinear Faà di Bruno trees and general obstruction packets remain outside this paper.

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1 The refinement problem

Paper I [1] resolved a generated object $x \in X$ by one complete idempotent family

$$P_\alpha P_\beta = \delta_{\alpha\beta} P_\alpha, \qquad \sum_{\alpha \in I} P_\alpha = I_X,$$

with exact reconstruction

$$x = \sum_{\alpha \in I} P_\alpha x.$$

A single decomposition, however, does not yet express scale. The present problem is to place these packet systems into a coherent tower

$$\mathcal{P}_0 \prec \mathcal{P}_1 \prec \mathcal{P}_2 \prec \cdots$$

so that every coarse packet is the exact sum of its fine descendants and every operator can be transported between levels without changing the underlying object.

The production language now becomes

$$\boxed{\text{seed} \to \text{address} \to \text{operate} \to \text{project} \to \text{refine} \to \text{weight} \to \text{complete} \to \text{reconstruct.}}$$

Three questions organize the paper.

(i) When is a fine packet family an exact refinement of a coarse family?

(ii) When can an operator inverse be reconstructed from weighted packet-transfer words?

(iii) Why does the dyadic sector sometimes require additional depth even when the corresponding odd-prime packets separate immediately?

1.1 Status of the statements

The paper uses the following labels.

- **Theorem/Proposition**: proved under the stated hypotheses.
- **Classical input**: an established algebraic fact used without claiming priority.
- **Worked example**: an exact realization of the definitions.
- **Numerical certificate**: a finite computational check of the implementation.
- **Boundary**: a claim intentionally not made.

The release freezes a theorem-completeness denominator of twenty-five items: the twenty-one numbered definitions, propositions, theorems, and corollaries in the main text; the exact Smith-phase identity in section 11.3; and the three algorithms in appendix B. Each item is mapped to one or more compiled Lean declarations. “Lean-complete” below always means complete relative to this revised, scope-explicit contract. Some rows still take application-specific certificates, and their boundaries are stated individually. It is not a claim that Lean proves the historical motivation, examples, floating-point residuals, or bibliographic priority. The formalization uses Lean 4 and Mathlib [2, 3].

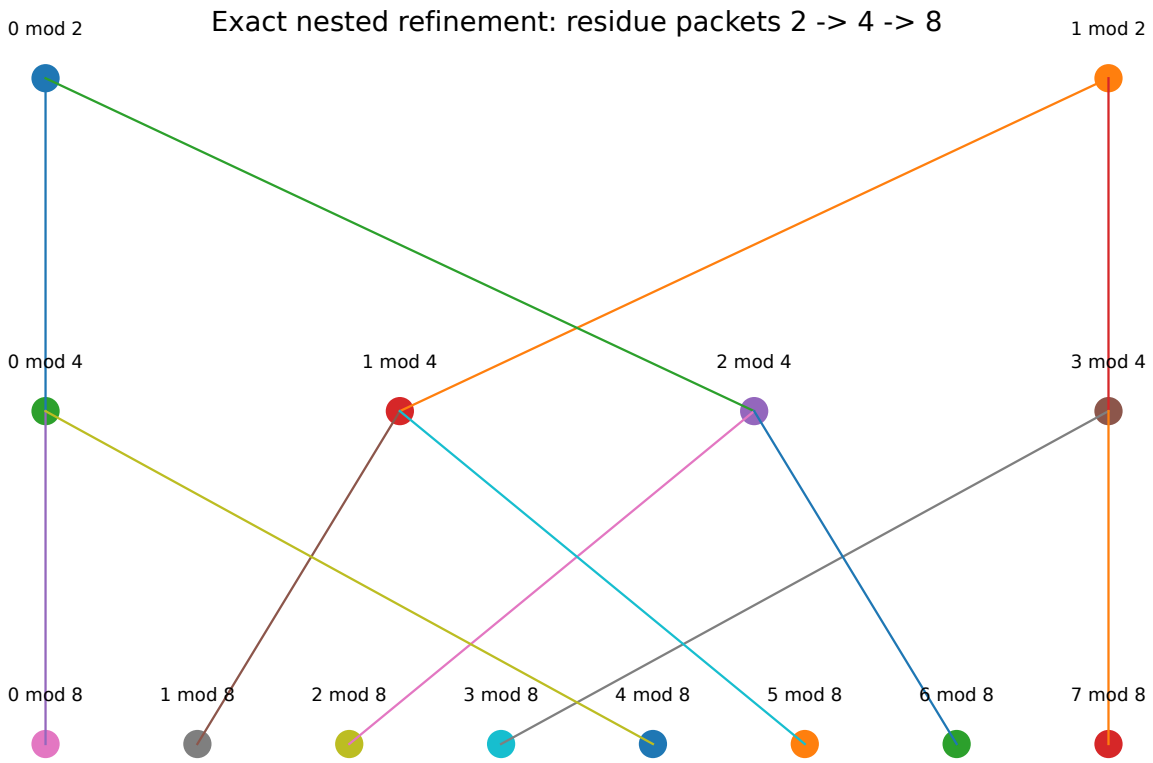


Figure 1: A nested residue hierarchy. Each residue modulo 2 is the union of two residues modulo 4, and each residue modulo 4 is the union of two residues modulo 8.

2 Nested packet systems

2.1 Refinement data

Let X be a module over a commutative ring R or a topological vector space over a field. At level ℓ, let

$$\mathcal{P}_\ell = \{P_{\ell,\alpha}\}_{\alpha \in I_\ell}$$

be a finite complete family of pairwise orthogonal idempotents.

Definition 2.1 (Refinement map)**.** A refinement map from level $\ell+1$ to level ℓ is a surjection

$$\rho_{\ell+1,\ell} : I_{\ell+1} \longrightarrow I_\ell$$

such that

$$\boxed{P_{\ell,\alpha} = \sum_{\rho_{\ell+1,\ell}(\beta)=\alpha} P_{\ell+1,\beta}}$$

for every $\alpha \in I_\ell$.

The relation $\beta \succ \alpha$ means that β is a fine child of α.

Theorem 2.2 (Nested refinement theorem)**.** *Suppose $\mathcal{P}_{\ell+1}$ refines $\mathcal{P}_\ell$. Then:*

(a) every fine packet lies inside its parent:

$$P_{\ell,\alpha}P_{\ell+1,\beta} = P_{\ell+1,\beta}P_{\ell,\alpha} = \begin{cases} P_{\ell+1,\beta}, & \beta \succ \alpha, \\ 0, & \textit{otherwise}; \end{cases}$$

(b) every coarse packet vector is reconstructed uniquely from its fine children;

(c) refinement is transitive: if $\mathcal{P}_{\ell+2}$ refines $\mathcal{P}_{\ell+1}$ and $\mathcal{P}_{\ell+1}$ refines $\mathcal{P}_\ell$, then $\mathcal{P}_{\ell+2}$ refines $\mathcal{P}_\ell$.

Proof. Multiply

$$P_{\ell,\alpha} = \sum_{\gamma \succ \alpha} P_{\ell+1,\gamma}$$

by $P_{\ell+1,\beta}$. Orthogonality at the fine level leaves exactly the term $P_{\ell+1,\beta}$ when $\beta \succ \alpha$, and zero otherwise. The reconstruction statement follows by applying the identity to x. For transitivity, substitute the child decomposition of each $P_{\ell+1,\beta}$ into the parent decomposition of $P_{\ell,\alpha}$ and collect all grandchildren whose composite parent is α. □

2.2 Analysis, synthesis, refinement, and coarsening

Define the packet-coordinate space

$$X_\ell^{\mathrm{pkt}} = \bigoplus_{\alpha \in I_\ell} P_{\ell,\alpha}X.$$

The analysis and synthesis maps are

$$\mathcal{A}_\ell x = (P_{\ell,\alpha}x)_\alpha, \qquad \mathcal{S}_\ell(x_\alpha)_\alpha = \sum_\alpha x_\alpha.$$

They satisfy

$$\mathcal{S}_\ell\mathcal{A}_\ell = I_X, \qquad \mathcal{A}_\ell\mathcal{S}_\ell = I_{X_\ell^{\mathrm{pkt}}}.$$

Definition 2.3 (Exact coordinate refinement)**.** Define

$$J_{\ell+1,\ell} = \mathcal{A}_{\ell+1}\mathcal{S}_\ell, \qquad C_{\ell,\ell+1} = \mathcal{A}_\ell\mathcal{S}_{\ell+1}.$$

The map J resolves coarse packet coordinates into fine coordinates, while C regroups fine coordinates by parent.

Proposition 2.4 (Perfect reconstruction across levels)**.** *The maps $J_{\ell+1,\ell}$ and $C_{\ell,\ell+1}$ are mutual inverses:*

$$C_{\ell,\ell+1}J_{\ell+1,\ell} = I, \qquad J_{\ell+1,\ell}C_{\ell,\ell+1} = I.$$

Proof. Use $\mathcal{S}_m\mathcal{A}_m = I_X$ at the intermediate level:

$$CJ = \mathcal{A}_\ell\mathcal{S}_{\ell+1}\mathcal{A}_{\ell+1}\mathcal{S}_\ell = \mathcal{A}_\ell\mathcal{S}_\ell = I.$$

The other identity is identical with the levels exchanged. □

These coordinate changes are coherent through every finite tower. If three complete packet systems are denoted by a, b, c, then

$$J_{c,b}J_{b,a} = J_{c,a}, \qquad J_{a,a} = I, \qquad J_{b,a}^{-1} = J_{a,b}.$$

These laws hold for arbitrary exact packet systems representing the same ambient module; the parent maps are needed for the componentwise parent–child formulas, not for coordinate coherence itself.

Moreover, let P_a, P_b be packet systems on X, let Q_a, Q_b be packet systems on Y, and let $T : X \to Y$ be a typed linear map. Its exact coordinate representation

$$\widehat{T}_{Q,P} = \mathcal{A}_Q T \mathcal{S}_P$$

preserves identities and typed composition, and the naturality square is

$$\boxed{J^Y_{Q_b,Q_a}\,\widehat{T}_{Q_a,P_a} = \widehat{T}_{Q_b,P_b}\,J^X_{P_b,P_a}.}$$

The identity, cocycle, composition, inverse, and naturality laws are all proved in Lean. The separate refinement hypotheses supply the local parent–child component formulas.

This result should not be confused with dimensional reduction. Both packet-coordinate spaces represent the full object. Refinement changes coordinates; it does not discard fine information.

2.3 Operator blocks under refinement

For $T \in \mathrm{End}(X)$, define

$$T^{(\ell)}_{\beta\alpha} = P_{\ell,\beta}TP_{\ell,\alpha}.$$

Theorem 2.5 (Transfer-block refinement)**.** *If $\mathcal{P}_{\ell+1}$ refines $\mathcal{P}_\ell$, then*

$$\boxed{T^{(\ell)}_{ba} = \sum_{\substack{\beta\succ b\\ \alpha\succ a}} T^{(\ell+1)}_{\beta\alpha}.}$$

Consequently the complete operator is reconstructed at every level:

$$T = \sum_{a,b} T^{(\ell)}_{ba} = \sum_{\alpha,\beta} T^{(\ell+1)}_{\beta\alpha}.$$

Proof. Expand both coarse projectors into their children:

$$P_{\ell,b}TP_{\ell,a} = \left(\sum_{\beta\succ b} P_{\ell+1,\beta}\right) T \left(\sum_{\alpha\succ a} P_{\ell+1,\alpha}\right).$$

Distribute the finite sums. □

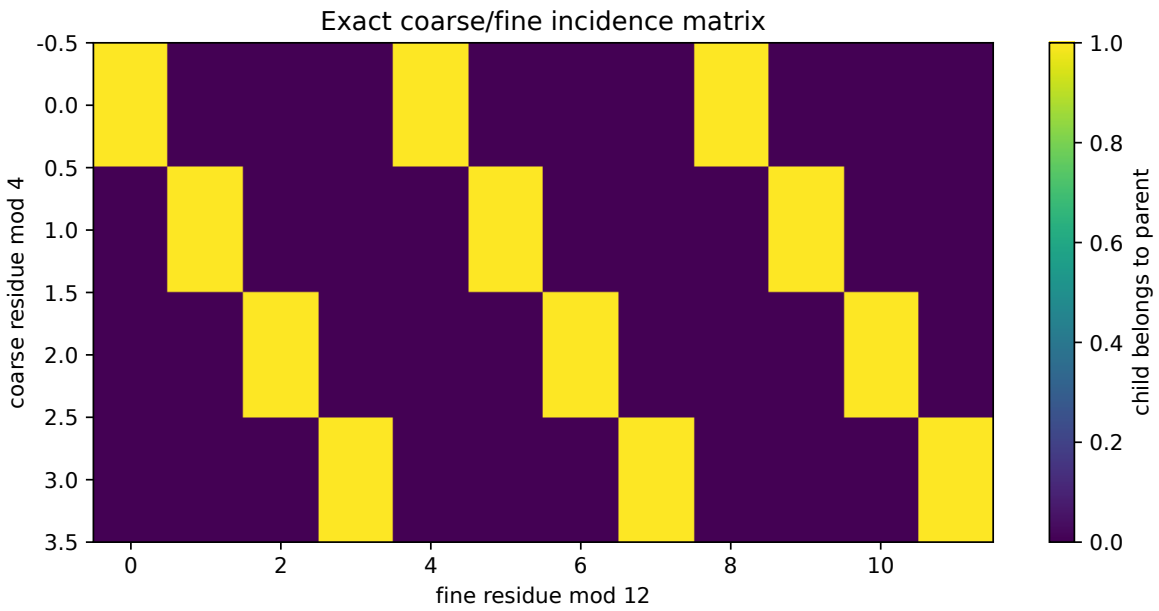


Figure 2: The incidence matrix $M_{r,s} = 1$ when the fine residue s belongs to the coarse residue r. Its rows implement exact fine-to-coarse packet aggregation.

3 Residue and polyphase selectors

3.1 Coefficient selectors over an arbitrary ring

Let

$$F(z) = \sum_{n \geq 0} a_n z^n$$

be a formal series over a ring R. For $N \geq 1$ and $0 \leq j < N$, define

$$\boxed{(P_{N,j}F)(z) = \sum_{n \equiv j \pmod N} a_n z^n.}$$

No division and no roots of unity are required.

Proposition 3.1 (Residue projector algebra)**.** *The coefficient selectors satisfy*

$$P_{N,i}P_{N,j} = \delta_{ij}P_{N,j}, \qquad \sum_{j=0}^{N-1} P_{N,j} = I.$$

Proof. Every coefficient address n belongs to exactly one residue class modulo N. Applying two selectors preserves the coefficient only when both residue labels agree with n. □

3.2 Finite Fourier realization

Let U act on coefficient sequences by

$$(UF)(z) = F(\omega z).$$

If $\omega^N = 1$, then $U^N = I$.

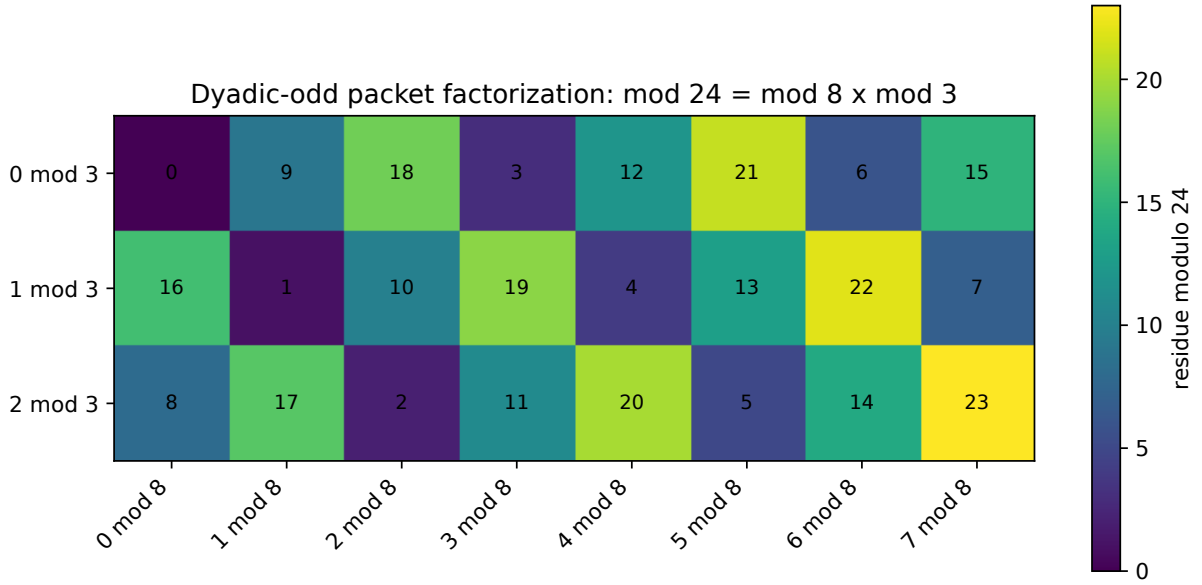


Figure 3: A composite residue packet is addressed by its dyadic and odd coordinates. The displayed mod-24 system is the product of mod 8 and mod 3.

Theorem 3.2 (Root-of-unity realization)**.** *Let R be a commutative semiring, let $N > 0$, and suppose $\omega, \omega^\vee, \eta \in R$ satisfy $\omega^N = 1$ and the finite $N \times N$ Fourier orthogonality table*

$$\eta \sum_{k=0}^{N-1} (\omega^\vee)^{jk} \omega^{rk} = \begin{cases} 1, & r = j, \\ 0, & r \neq j, \end{cases} \qquad 0 \le j, r < N.$$

Then

$$\boxed{P_{N,j} = \eta \sum_{k=0}^{N-1} (\omega^\vee)^{jk} U^k.}$$

In particular, if R is a field, $N > 0$, and ω is a primitive Nth root of unity, the certificate holds with $\omega^\vee = \omega^{-1}$ and $\eta = N^{-1}$.

Proof. On the monomial z^n, the right-hand side acts by

$$\eta \sum_{k=0}^{N-1} (\omega^\vee)^{jk} \omega^{nk}.$$

The identity $\omega^N = 1$ reduces this expression to the table entry indexed by $r = n \bmod N$. The finite certificate therefore gives exactly the coefficient mask defining $P_{N,j}$. Over a field, the primitive-root geometric sum supplies the certificate. □

The field or orthogonality condition is essential. A primitive element of order N in a ring with zero divisors does not by itself force the off-diagonal geometric sums to vanish. For example, in $\mathbb{F}_3 \times \mathbb{F}_3$ the element $(-1, 1)$ has order two, but the second component of its nontrivial character sum is nonzero. The division-free coefficient selector remains valid without any Fourier certificate. This distinction becomes decisive at the dyadic prime.

3.3 Nested residue refinement

Theorem 3.3 (Polyphase refinement)**.** *If $M = Nq$ with $N, q > 0$, then*

$$\boxed{P_{N,j} = \sum_{\substack{0 \le s < M \\ s \equiv j \pmod N}} P_{M,s}.}$$

Proof. An address n satisfies $n \equiv j \pmod N$ exactly when its unique residue $s = n \pmod M$ satisfies $s \equiv j \pmod N$. □

Thus the familiar $2 \to 4 \to 8$ hierarchy is exact, not approximate:

$$P_{2,0} = P_{4,0} + P_{4,2} = P_{8,0} + P_{8,2} + P_{8,4} + P_{8,6},$$

$$P_{2,1} = P_{4,1} + P_{4,3} = P_{8,1} + P_{8,3} + P_{8,5} + P_{8,7}.$$

4 Dyadic–odd factorization

Let

$$N = 2^a M, \qquad M \text{ odd}.$$

The two moduli are coprime.

Theorem 4.1 (Dyadic–odd packet factorization)**.** *For every residue $j \pmod N$, let*

$$j_2 = j \pmod{2^a}, \qquad j_o = j \pmod M.$$

Then

$$\boxed{P_{N,j} = P_{2^a,j_2} P_{M,j_o} = P_{M,j_o} P_{2^a,j_2}.}$$

Conversely, every pair (j_2, j_o) labels one and only one packet modulo N.

Proof. A coefficient address n survives the product on the right exactly when

$$n \equiv j_2 \pmod{2^a}, \qquad n \equiv j_o \pmod M.$$

The Chinese remainder theorem gives one residue $j \pmod{2^a M}$ satisfying both conditions. □

This theorem justifies separating the coordinate system into

$$\boxed{\text{dyadic phase address} \times \text{odd phase address}.}$$

It does not assert that their arithmetic lifting theories are identical. The coordinate factorization is exact, while the integral projector weights can still differ sharply.

Example 4.2 (Mod 12)**.** Since $12 = 4 \cdot 3$, every residue is identified by

$$(j \bmod 4, j \bmod 3).$$

For instance, the pair $(3, 1)$ corresponds to $j = 7 \pmod{12}$, so

$$P_{12,7} = P_{4,3} P_{3,1}.$$

5 Weighted transfer words

5.1 Filtered operator algebras

Let $\mathcal{O}$ be a complete discrete valuation ring with uniformizer π, fraction field K, and filtration

$$F^m\mathcal{B} = \pi^m\mathcal{B}$$

on an $\mathcal{O}$-algebra $\mathcal{B}$. Suppose

$$T = D + H,$$

where D is invertible and the perturbation H raises filtration. The symbol K is reserved for the fraction field.

A packet decomposition resolves $H = \sum_{e\in E} H_e$ and assigns a certified filtration gain

$$D^{-1}H_e \in F^{w(e)}\mathcal{B}, \qquad w(e) \in \mathbb{Z}_{>0}.$$

Each e is a directed transfer edge. For an ordered transfer word

$$\gamma = (e_1, \dots, e_r),$$

define

$$W(\gamma) = \sum_{j=1}^{r} w(e_j), \qquad A_\gamma = A_{e_1} \cdots A_{e_r}, \qquad A_e = D^{-1}H_e.$$

Theorem 5.1 (Certified weighted inverse)**.** *Let R be a possibly noncommutative ring, let D^{-1} be a supplied two-sided inverse of D, and put $A = D^{-1}H$.*

(a) If $A^r = 0$, then $D + H$ has the two-sided inverse

$$\boxed{(D+H)^{-1} = \left(\sum_{k=0}^{r-1}(-A)^k\right) D^{-1}.}$$

(b) If R is a complete normed ring and $\|A\| < 1$, then $\sum_{k\geq 0}(-A)^k$ converges and

$$\boxed{(D+H)^{-1} = \left(\sum_{k\geq 0}(-A)^k\right) D^{-1}.}$$

Proof. Factor

$$T = D(I + A), \qquad A = D^{-1}H.$$

For a finite cutoff r, multiplication on either side gives the noncommutative telescoping identity

$$(I+A)\sum_{k=0}^{r-1}(-A)^k = \left(\sum_{k=0}^{r-1}(-A)^k\right)(I+A) = 1 - (-A)^r.$$

This proves (a). In (b), the strict norm bound makes the geometric series summable. Continuity of multiplication passes the same identities to the limit. Multiplication by D^{-1} gives the displayed inverse in both orders. □

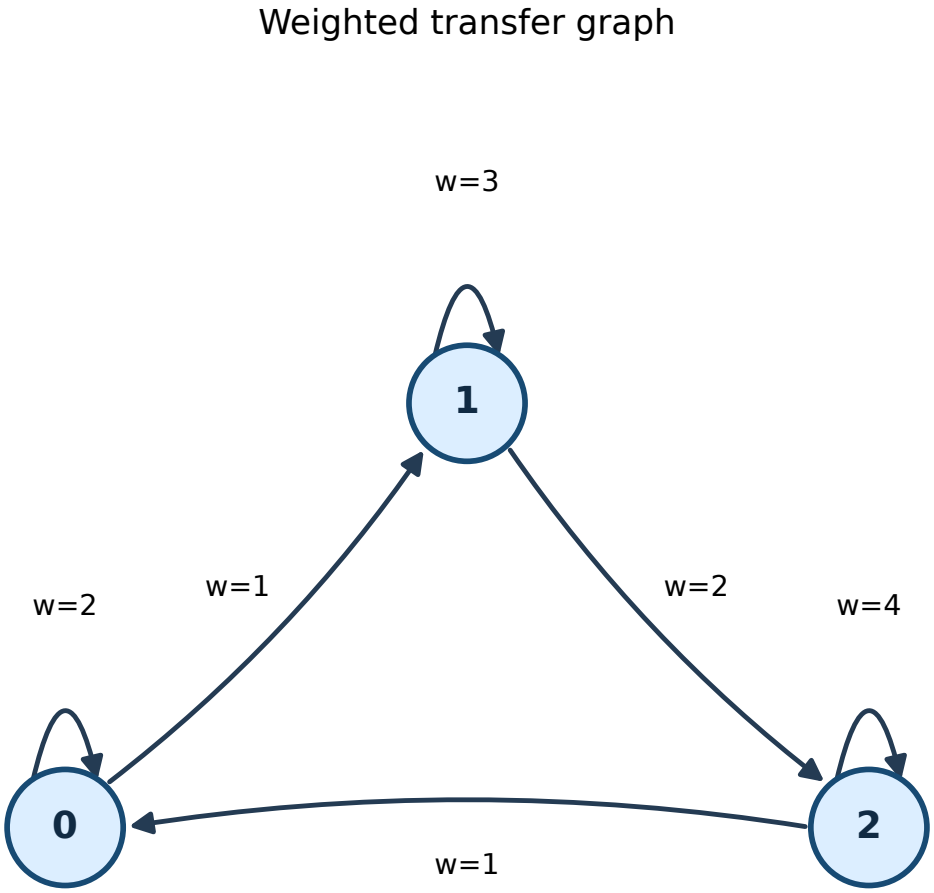


Figure 4: A finite packet-transfer graph. Edge weights record filtration gain rather than merely the number of operator compositions.

Corollary 5.2 (Finite precision)**.** *Let $F^\bullet R$ be a multiplicative decreasing filtration and let $q_L : R \to R_L$ be a ring homomorphism that kills $F^L R$. Suppose*

$$H = \sum_{e \in E} H_e, \qquad D^{-1} H_e \in F^{w(e)} R, \qquad w(e) \geq 1,$$

with E finite. Then $q_L(D^{-1}H)^L = 0$, the finite sum in theorem 5.1(a) is a two-sided inverse after applying q_L, and its word expansion may discard every word of weight at least L. More explicitly, for a length cutoff N, define

$$S^{\mathrm{pr}}_{L,N} = \sum_{n=0}^{N-1} (-1)^n \sum_{\substack{\gamma \in E^n \\ W(\gamma) < L}} A_\gamma.$$

Lean proves directly that $q_L(S^{\mathrm{pr}}_{L,N})$ equals the mapped geometric inverse whenever $q_L(A^N) = 0$, and hence is a two-sided inverse of $q_L(I + A)$, where $A = \sum_e A_e = D^{-1}H$. With the supplied two-sided inverse D^{-1} of D, this gives the explicit two-sided inverse

$$\boxed{q_L\left(S^{\mathrm{pr}}_{L,N} D^{-1}\right)} \quad \textit{of} \quad q_L(D + H).$$

If every edge weight is at least m and $L \leq mN$, then the nilpotence condition follows from the filtration data because $m|\gamma| \leq W(\gamma)$. For $m > 0$, the executable choice $N = \lceil L/m \rceil$ satisfies $L \leq mN$; this elementary ceiling inequality is also proved in Lean. The choice $m = 1, N = L$ gives the standard cutoff. In the DVR example $q_L : R \to R/\pi^L R$, this reads

$$q_L(T)^{-1} = q_L\left(\sum_{W(\gamma) < L} (-1)^{|\gamma|} A_\gamma D^{-1} \right).$$

Here $A_\gamma = (D^{-1}H_{e_1}) \cdots (D^{-1}H_{e_r})$ in the written order. Multiplicativity places it in $F^{W(\gamma)}R$, so the quotient kills it when $W(\gamma) \geq L$. Positive weights also imply $|\gamma| \leq W(\gamma)$, making the retained search finite.

No unrestricted convergence assertion for every complete filtered algebra is used in the formal contract. Such an assertion needs a separately specified topology and completeness theorem. The release proves the finite-quotient result above and the complete-normed contraction specialization.

6 Transfer-zeta functions

Let G be a finite directed graph with positive integer edge weights. Define the polynomial adjacency matrix

$$A(t)_{ij} = \sum_{e : i \to j} t^{w(e)},$$

where parallel edges are distinct elements of the finite edge type and are therefore counted with their full multiplicity.

For fixed start and target vertices s, t, define

$$Z_{s,t}(t) = \sum_{n \geq 0} a_n(s,t) t^n,$$

where $a_n(s,t)$ counts directed walks from s to t with total weight n. Vertices and edges may repeat. Thus "path zeta" is retained as the name of the series and Lean API, but no simple-path restriction is intended.

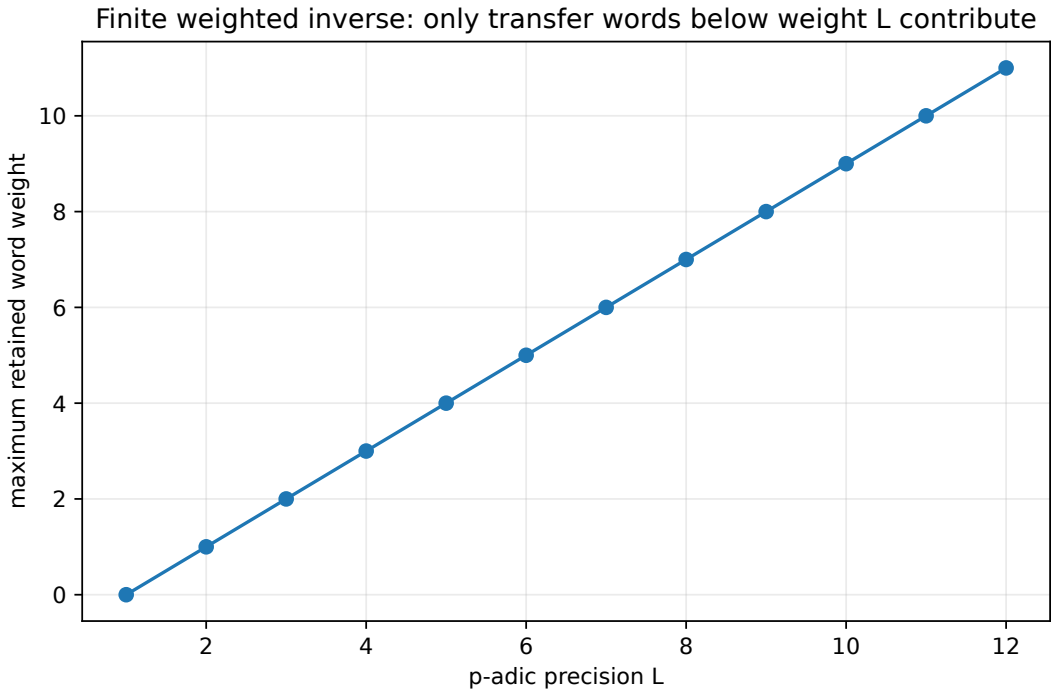


Figure 5: At precision L, weighted inversion requires only transfer words whose cumulative weight is below L.

Theorem 6.1 (Automatic finite-graph transfer-zeta rationality)**.** *Assume the vertex and edge sets are finite and every edge has positive integer weight. For every n, exact enumeration of edge tuples of lengths at most n, followed by endpoint and total-weight filtering, returns $a_n(s,t)$. These coefficients obey the first-edge recurrence*

$$a_n(s,t) = \mathbf{1}_{s=t,\,n=0} + \sum_{\substack{e:\,\mathrm{source}(e)=s \\ w(e)\le n}} a_{n-w(e)}(\mathrm{target}(e),t).$$

Let

$$T(t) = I - A(t), \qquad Q(t) = \det T(t), \qquad P_{s,t}(t) = \mathrm{adj}(T(t))_{s,t}.$$

Then $Q(0) = 1$, hence $Q \neq 0$ and Q is a unit in $\mathbb{Z}[[t]]$, and the independently enumerated walk series satisfies

$$\boxed{Z_{s,t}(t) = Q(t)^{-1}P_{s,t}(t), \qquad Q(t)Z_{s,t}(t) = P_{s,t}(t)} \quad \text{in } \mathbb{Z}[[t]].$$

Proof. Every edge has weight at least one, so a word of weight n has length at most n. The enumerator therefore searches a finite set containing every contributing walk, and its filters are exactly the defining predicates for a_n. Removing the first edge gives the displayed disjoint decomposition into the empty diagonal walk and the first-edge tails. In matrix form this is

$$Z = I + AZ, \qquad (I - A)Z = I.$$

Positive weights give $A(0) = 0$, so $T(0) = I$ and $Q(0) = \det(I) = 1$. The two adjugate identities

$$T\,\mathrm{adj}(T) = QI, \qquad \mathrm{adj}(T)T = QI$$

remain valid after inclusion into $\mathbb{Z}[[t]]$. Multiplying the adjugate by the formal-series inverse of Q constructs a two-sided inverse of T. Uniqueness of the inverse identifies that construction with the independently enumerated matrix Z, yielding the entrywise formulas. □

This is the finite transfer-matrix method of [6], but here its determinant, adjugate, normalization, first-edge recurrence, and identification with the enumerated series are all formalized. The older API accepting an independently supplied normalized coefficient identity $QZ = P$ is retained for external certificates; it is no longer needed to obtain rationality for the finite graph defined above. The Python companion performs an exact symbolic construction as an independent executable cross-check.

Corollary 6.2 (Linear recurrence)**.** *With the canonical polynomials from theorem 6.1, write*

$$Q(t) = \sum_k q_k t^k, \qquad Z_{s,t}(t) = \sum_{n\geq 0} a_n(s,t) t^n.$$

Then the walk counts satisfy the constant-coefficient recurrence determined by Q after the finite initial range contributed by $P_{s,t}$:

$$\boxed{\sum_{k=0}^{n} q_k a_{n-k}(s,t) = 0 \qquad (n > \deg P_{s,t}).}$$

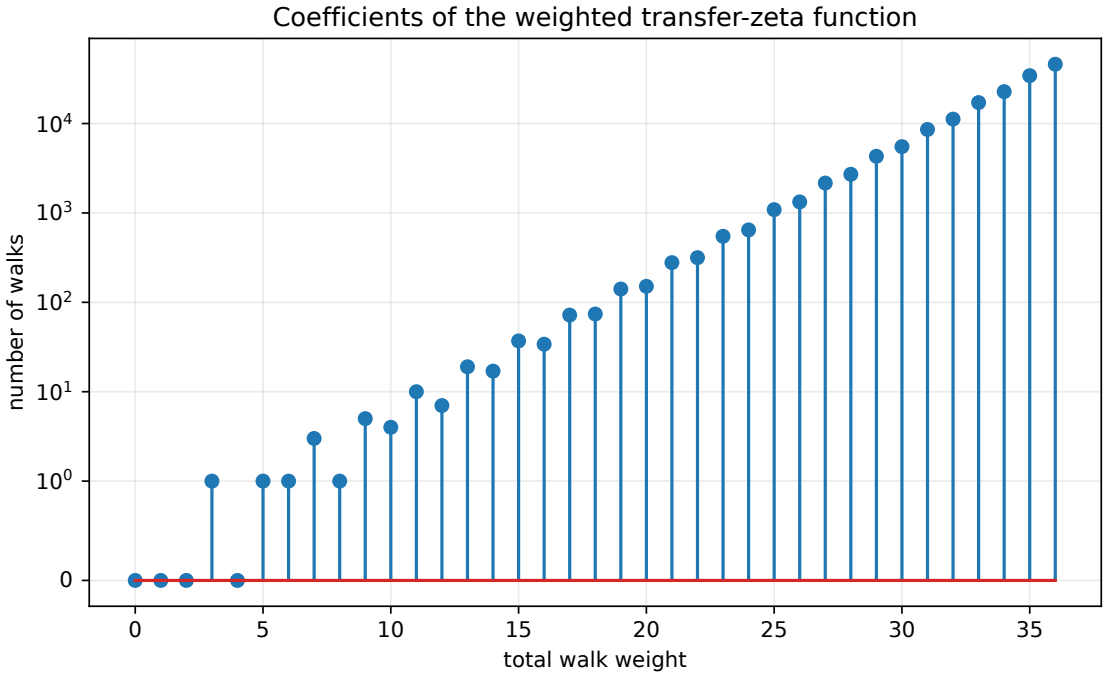


Figure 6: The walk counts for one weighted graph. The exact generating function is rational, so these counts satisfy a finite linear recurrence.

Example 6.3 (A computed transfer zeta)**.** For the graph used in the numerical companion, the walk series from packet 0 to packet 2 is

$$Z_{0,2}(t) = \frac{t^3}{1 - t^2 - t^3 - 2t^4 + t^5 + t^6 + t^7 - t^9}.$$

The first nonzero coefficients are

$$1, 1, 1, 3, 1, 5, 4, 10, 7, 19, 17, 37, \ldots$$

at their corresponding weights. Direct enumeration through weight 30 agrees coefficient by coefficient with the rational expansion.

7 Integral spectral packets

7.1 Bézout projectors

Let $\mathcal{O}$ be a DVR, $K = \mathrm{Frac}(\mathcal{O})$, and $A \in \mathrm{End}_{\mathcal{O}}(\Lambda)$ for a finite free lattice Λ. Suppose the characteristic polynomial factors over $\mathcal{O}$ as

$$\chi_A(x) = f(x)g(x),$$

where f and g are coprime over $K[x]$.

Theorem 7.1 (Evaluated Bézout spectral projector)**.** *Let $F, G, U, V \in \mathrm{End}_K(\Lambda \otimes_{\mathcal{O}} K)$ and let $r \in K^\times$. Suppose*

$$UF + VG = rI, \qquad FG = 0, \qquad FG = GF, \qquad GU = UG.$$

Put $N = VG$. Then

$$\boxed{N^2 = rN.}$$

Consequently the operator

$$\boxed{E = r^{-1}VG}$$

is idempotent, acts as the identity on $\ker F$, and acts as zero on $\ker G$. If V and G preserve Λ, then

$$rE = VG \in \mathrm{End}_{\mathcal{O}}(\Lambda).$$

Proof. The Bézout identity gives $N = rI - UF$. The two stated commutation relations and $FG = 0$ give

$$N(UF) = VGUF = VUGF = VUFG = 0.$$

Therefore

$$N^2 = N(rI - UF) = rN.$$

Since $r \neq 0$, $E^2 = r^{-2}N^2 = E$. If $Fx = 0$, the evaluated Bézout identity gives $VGx = rx$, hence $Ex = x$. If $Gx = 0$, then $Ex = 0$. The lattice assertion follows directly from $rE = VG$. □

The denominator-free equality $N^2 = rN$ uses neither $r \neq 0$ nor the other pairwise commutation relations that hold in the polynomial application. The normalization to E uses $r \neq 0$; action on $\ker F$ uses only the Bézout identity and $r \neq 0$, while vanishing on $\ker G$ is immediate from $N = VG$. The bundled Lean certificate records all pairwise commutations as convenient evaluated polynomial data, but the minimal theorems expose only the hypotheses they consume.

For the polynomial application, one supplies $F = f(A)$, $G = g(A)$, $U = u(A)$, and $V = v(A)$. A Bézout or resultant identity $uf + vg = r$ evaluates to the first certificate, while $(fg)(A) = 0$ supplies the second; when $fg = \chi_A$, this last equality is Cayley–Hamilton. Standard Bézout, resultant, and DVR facts are recalled from [5, 4]. The formal theorem begins with the evaluated certificate. It does not infer exact action merely from nilpotence of $f(A)$ or $g(A)$, which would be insufficient.

Definition 7.2 (Separation weight)**.** For a rational projector E and a specified $\mathcal{O}$-lattice Λ, provided at least one clearing exponent exists, define

$$s_{\pi,\Lambda}(E) = \min\{s \geq 0 : \pi^s E(\Lambda) \subseteq \Lambda\}.$$

If the evaluated certificate arises from integral v, g and $r = u\pi^s$ with u^{-1} preserving the lattice, then theorem 7.1 gives the upper bound

$$s_{\pi,\Lambda}(E) \leq s = \mathrm{val}_\pi(r).$$

Indeed,

$$\boxed{\pi^s E = u^{-1} V G}$$

preserves Λ. The minimal Lean theorem needs only the lattice action of u^{-1}; its bundled lattice-unit wrapper records both u and u^{-1}, rather than treating every nonzero fraction-field scalar as an integral unit. Existence of a clearing exponent is an explicit hypothesis, not a consequence of the notation. Once one exists, the least one is obtained by well-ordering. Clearing exponents are upward closed provided multiplication by π preserves the lattice.

7.2 Precision delay

Theorem 7.3 (Projector precision delay)**.** *Let E be a rational projector with $\pi^s E(\Lambda) \subseteq \Lambda$. If*

$$x \equiv y \pmod{\pi^{m+s}\Lambda},$$

then

$$Ex \equiv Ey \pmod{\pi^m \Lambda}.$$

Thus $m + s$ input digits suffice to determine m output digits of the projected packet.

Proof. Write $x - y = \pi^{m+s} z$. Since $\pi^s E(z) \in \Lambda$,

$$E(x - y) = \pi^m(\pi^s E(z)) \in \pi^m \Lambda.$$

□

The theorem makes delayed separation quantitative. A packet with weight s is not absent; it requires s additional digits before it can be reconstructed to the same output precision.

8 The exceptional dyadic sign split

Consider the integral involution

$$A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad A^2 = I.$$

Over a field of characteristic different from two, its sign projectors are

$$E_+ = \frac{I + A}{2} = \begin{pmatrix} 1 & 1/2 \\ 0 & 0 \end{pmatrix},$$

$$E_- = \frac{I - A}{2} = \begin{pmatrix} 0 & -1/2 \\ 0 & 1 \end{pmatrix}.$$

They are orthogonal idempotents and sum to I.

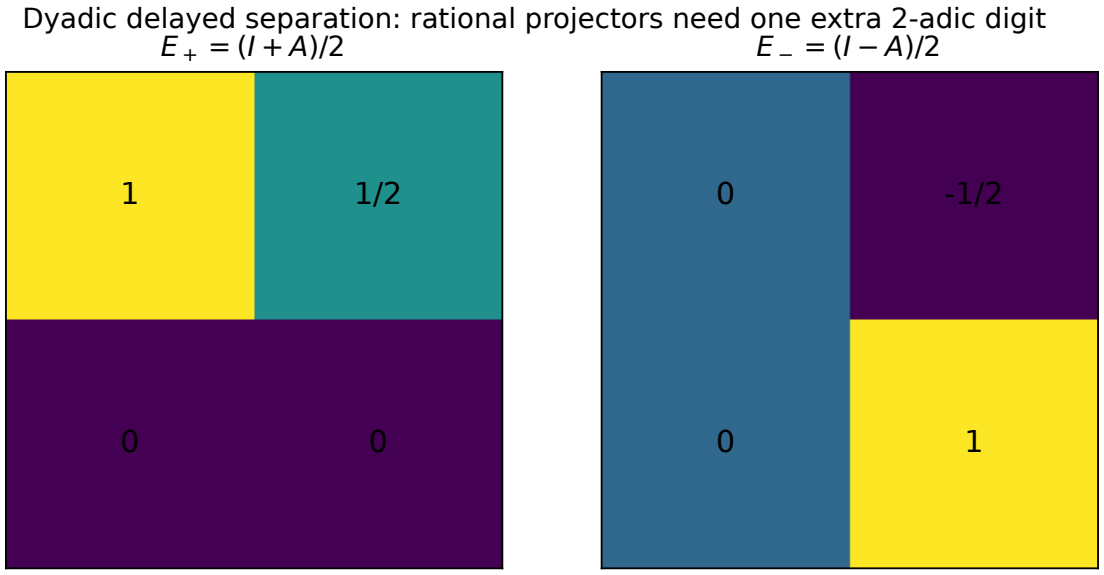


Figure 7: The rational sign projectors for an integral involution. Their denominator has 2-adic weight one.

Proposition 8.1 (Odd-adic versus dyadic behavior)**.** *For every prime $p \neq 2$, every entry of E_+ and E_- is p-integral, and each projector has exact local denominator weight zero. At $p = 2$, each projector has exact local denominator weight one; multiplication by 2 makes it integral, while the unscaled matrix is not integral:*

$$s_{2,\mathbb{Z}^2}(E_+) = s_{2,\mathbb{Z}^2}(E_-) = 1.$$

The characteristic factors $x - 1$ and $x + 1$ have resultant 2. Its 2-adic valuation is one, exactly the two local projector weights above, and its p-adic valuation is zero at every prime $p \neq 2$.

Proof. The reduced denominators of the displayed entries are 1 or 2, which proves the local integrality and minimality statements directly. The resultant is

$$\mathrm{Res}(x - 1, x + 1) = 2.$$

Thus $v_2(2) = 1$ and $v_p(2) = 0$ for primes $p \neq 2$. □

The two characteristic factors $x - 1$ and $x + 1$ have the same reduction in $\mathbb{F}_2[x]$. The rational matrices E_+ and E_- themselves cannot be reduced modulo 2, because their entries contain $1/2$. Consequently the integral special fibre does not inherit their separate idempotent splitting, even though the two projectors remain distinct over $\mathbb{Q}_2$. This is the simplest model of delayed dyadic separation.

8.1 Why coefficient packets remain available

The failure of the character formula

$$\frac{1}{2}(I \pm A)$$

to be integral does not invalidate coefficient selectors such as even and odd residue extraction. It shows that two distinct constructions must be distinguished:

integral address selector	$\neq$	character-average spectral projector

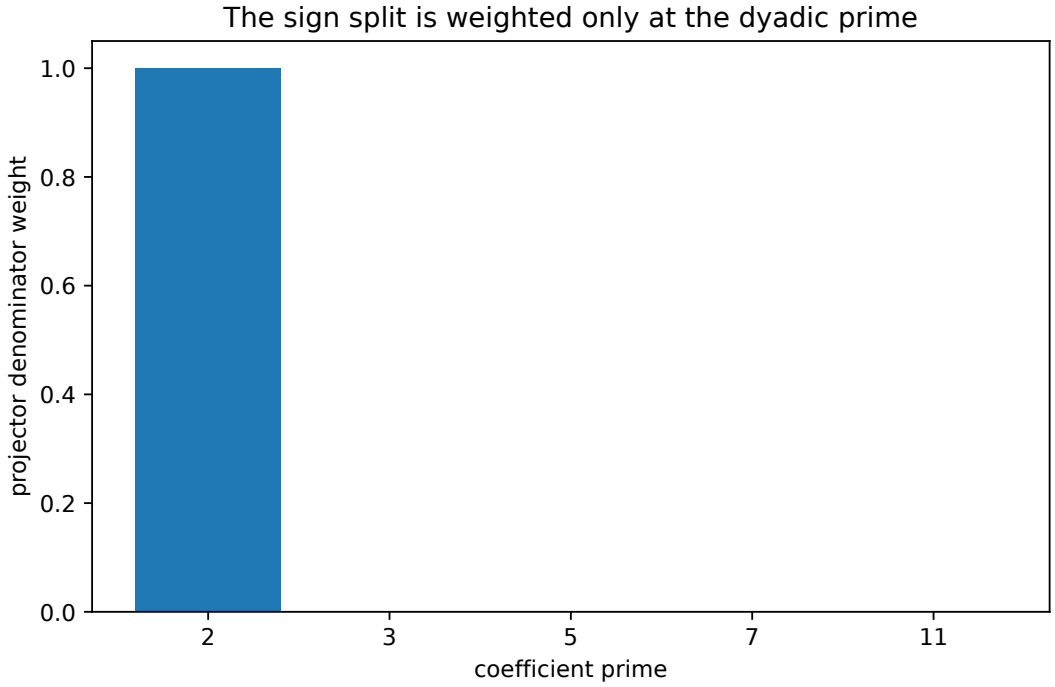


Figure 8: The denominator weight of the sign projector is nonzero only at the prime two.

when the group order is not a unit. The first can remain exact over $\mathbb{Z}_2$; the second can acquire a denominator weight.

9 Smith weights and multivariable lifting

Let M be a matrix over a DVR. Smith normal form gives invertible matrices U, V such that

$$UMV = \mathrm{diag}(\pi^{s_1}u_1, \ldots, \pi^{s_r}u_r, 0, \ldots, 0),$$

where each u_i is a unit and

$$0 \le s_1 \le \cdots \le s_r.$$

The exponents s_i are directional lifting weights.

Theorem 9.1 (Concrete Smith-weighted congruence solve)**.** *Let $p > 1$ and consider the diagonal system over $(\mathbb{Z}/(p^L)\mathbb{Z})^d$*

$$\mathrm{diag}(p^{s_1}, \ldots, p^{s_d})x = b \pmod{p^L}.$$

The system is solvable if and only if

$$p^{\min(s_i,L)} \mid (b_i)_{\mathrm{val}}$$

for every i, where $(b_i)_{\mathrm{val}}$ is the canonical natural representative. Conditional on this admissibility condition, the number of solutions is

$$\boxed{p^{\sum_{i=1}^{d}\min(s_i,L)}}.$$

If any coordinate is inadmissible, the solution set is empty and its cardinality is zero.

Proof. Lean proves the one-coordinate arithmetic directly over $\mathbb{Z}/p^L\mathbb{Z}$ for every $p > 1$:

$$p^s x = b \pmod{p^L} \quad \Longleftrightarrow \quad p^{\min(s,L)} \mid b_{\mathrm{val}},$$

where b_{val} is the canonical natural representative, and every admissible fibre has exactly $p^{\min(s,L)}$ elements. The coordinates are independent, so the product equivalence gives the displayed global count. If one coordinate fibre is empty, the global fibre is empty. Lean proves both the global solvability equivalence and the conditional count, including cardinality zero on the inadmissible branch. □

For an arbitrary DVR $\mathcal{O}$ with residue-field cardinality q, the analogous formula

$$q^{\sum_i \min(s_i,L)}$$

is a classical consequence of the corresponding one-coordinate solvability and fibre-cardinality facts. In the Lean development this general form is *certificate-explicit*: those coordinate facts may be supplied to the abstract product API, but they are not derived from a formalized general-DVR structure theorem. Construction and diagonalization of Smith normal form for an arbitrary DVR matrix also remain classical input; see [4, 5].

If a Smith form is rank deficient, a literal zero diagonal factor requires the corresponding target coordinate to vanish modulo p^L and then contributes all p^L possible coordinate values. In the concrete quotient it can equivalently be represented by any weight $s_i \geq L$, since $p^{s_i} = 0$ in $\mathbb{Z}/(p^L)\mathbb{Z}$; the theorem then gives exactly that condition and factor. The numbered theorem begins after the diagonal data have been supplied and does not construct the Smith form.

Example 9.2. At $p = 2$, depth $L = 6$, and weights $(0, 1, 3)$, the right-hand side $(1, 2, 8)$ is admissible. The coordinate kernels have sizes

$$1, 2, 8,$$

so the system has 16 solutions. Replacing 8 by 4 in the final coordinate violates the divisibility condition and makes the system unsolvable.

This theorem is the multivariable version of precision delay. A coordinate with weight s_i contributes both a solvability threshold and s_i units of branching until the ambient depth truncates it.

10 Linear refinement defects

Exact coordinate refinement does not guarantee that an independently chosen coarse operator intertwines with a fine operator. Let $C : X_f \to X_c$ be any linear coarsening map and let $T_f \in \mathrm{End}(X_f)$ and $T_c \in \mathrm{End}(X_c)$. Define

$$\boxed{\Delta = CT_f - T_c C.}$$

Theorem 10.1 (Power-defect identity)**.** *For every $n \geq 1$,*

$$\boxed{CT_f^n - T_c^n C = \sum_{k=0}^{n-1} T_c^{n-1-k} \Delta T_f^k.}$$

Proof. The identity is immediate for $n = 1$. For the general case, insert and subtract $T_c C T_f^{n-1}$:

$$CT_f^n - T_c^n C = (CT_f - T_c C)T_f^{n-1} + T_c(CT_f^{n-1} - T_c^{n-1} C),$$

and apply induction. □

Corollary 10.2 (Norm bound)**.** *For a submultiplicative operator norm,*

$$\|CT_f^n - T_c^n C\| \le \|\Delta\| \sum_{k=0}^{n-1} \|T_c\|^{n-1-k} \|T_f\|^k.$$

If $\|T_f\|, \|T_c\| \le \rho$*, then*

$$\boxed{\|CT_f^n - T_c^n C\| \le n\rho^{n-1}\|\Delta\|.}$$

Theorem 10.3 (Resolvent-defect identity)**.** *If* $z - T_f$ *and* $z - T_c$ *are invertible, then*

$$\boxed{C(z - T_f)^{-1} - (z - T_c)^{-1}C = (z - T_c)^{-1}\Delta(z - T_f)^{-1}.}$$

Proof. Multiply the left-hand side on the left by $z - T_c$ and on the right by $z - T_f$:

$$(z - T_c)C - C(z - T_f) = CT_f - T_c C = \Delta.$$

Then invert the two exterior factors. □

For continuous linear maps, the same identity gives the certified norm bound

$$\boxed{\|CR_f - R_c C\| \le \|R_c\|\,\|\Delta\|\,\|R_f\|,}$$

from a left inverse witness for the coarse scalar shift and a right inverse witness for the fine scalar shift. These one-sided witnesses are sufficient for the displayed identity, although they need not by themselves define two-sided resolvents. If $\Delta = 0$, Lean also proves, for every finite polynomial p, the exact compatibility

$$\boxed{C\,p(T_f) = p(T_c)\,C,}$$

in both the algebraic and continuous-linear-map settings. Thus all powers, polynomial actions, and the explicitly witnessed resolvents intertwine with coarse reconstruction. This is the linear prototype of the more general refinement-defect theory reserved for Paper III.

11 Worked realizations in the common operator language

11.1 Signal polyphase coordinates

Let $x = (x_n)_{n\in\mathbb{Z}} \in c_{00}(\mathbb{Z}; R)$ and let the seed be δ_0 under the bilateral shifts $(S^r\delta_0)_n = \delta_{n,r}$. This makes the algebraic span and every displayed sum finite. The N packets are

$$x_j[m] = x[Nm + j].$$

Synthesis interleaves these sequences. A finite-impulse-response linear time-invariant filter becomes an $N \times N$ matrix of transfer blocks. Refining from N to M with $N \mid M$ splits each coarse phase into its descendants without changing the reconstructed signal. The same formulas extend to standard sequence completions when the synthesis series converges; this is the usual polyphase viewpoint from multirate filter banks [7].

11.2 Transfer-block sparsity

A banded operator need not remain banded after residue reordering, but its nonzero couplings become a structured packet graph. With the convention $(S_r x)_n = x_{n-r}$, the only active transfer is

$$P_{N,j+r} S_r P_{N,j},$$

with indices taken modulo N. More general finite stencils produce a finite set of active phase differences.

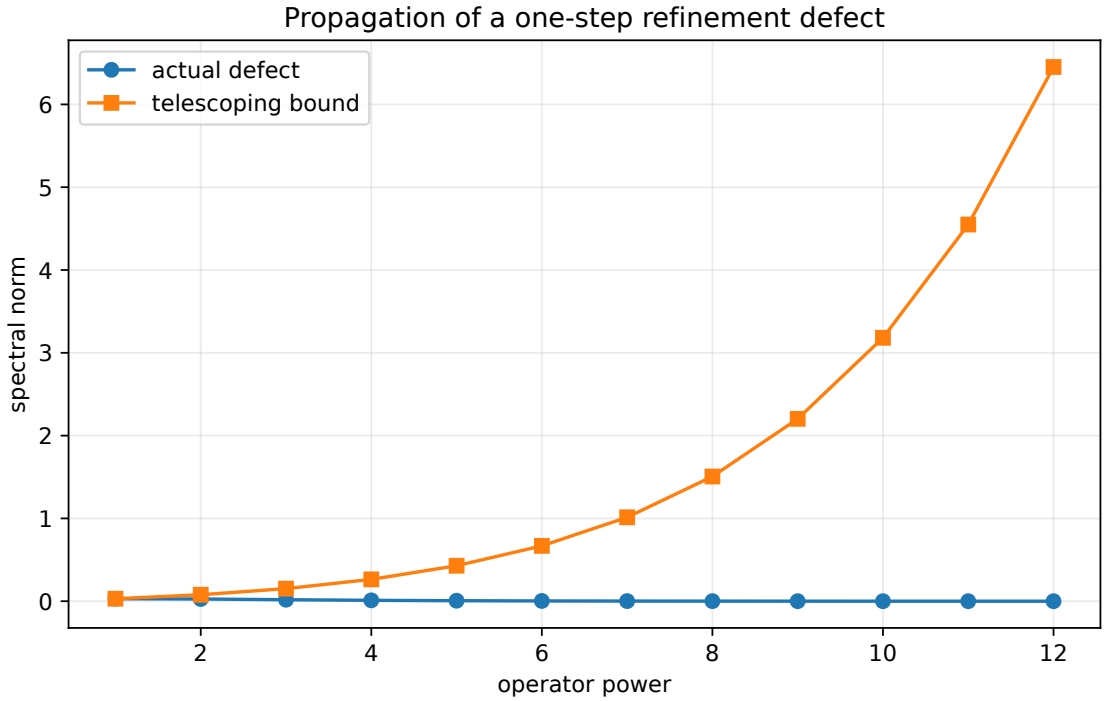


Figure 9: A controlled one-step defect and its telescoping norm bound under repeated operator application.

11.3 A short Smith-coordinate realization

Normalize an impedance by $z = Z/Z_0$ and write

$$\Gamma = \frac{z-1}{z+1}.$$

A phase action on the reflection disk is

$$\Gamma \mapsto \omega_N^k \Gamma.$$

Pulling it back gives

$$T_{N,k}(z) = \frac{(1+\omega_N^k)z + (1-\omega_N^k)}{(1-\omega_N^k)z + (1+\omega_N^k)}.$$

For $N = 2$ and $k = 1$,

$$T_{2,1}(z) = \frac{1}{z}, \qquad z \neq 0, -1.$$

Thus the nontrivial first-level phase action exchanges normalized impedance and admittance. Calling these values "packets" is geometric terminology for the sampled orbit, not an assertion that the two values form linear idempotent ranges. Higher 2-power refinements sample the same reflection circle at progressively finer electrical phases; compare the standard Smith chart conventions in [8].

The symmetric weight

$$w_{\mathrm{Smith}}(\Gamma) = \frac{1}{|1-\Gamma|^2} + \frac{1}{|1+\Gamma|^2}$$

indicates where small reflection-coordinate errors produce large impedance or admittance errors.

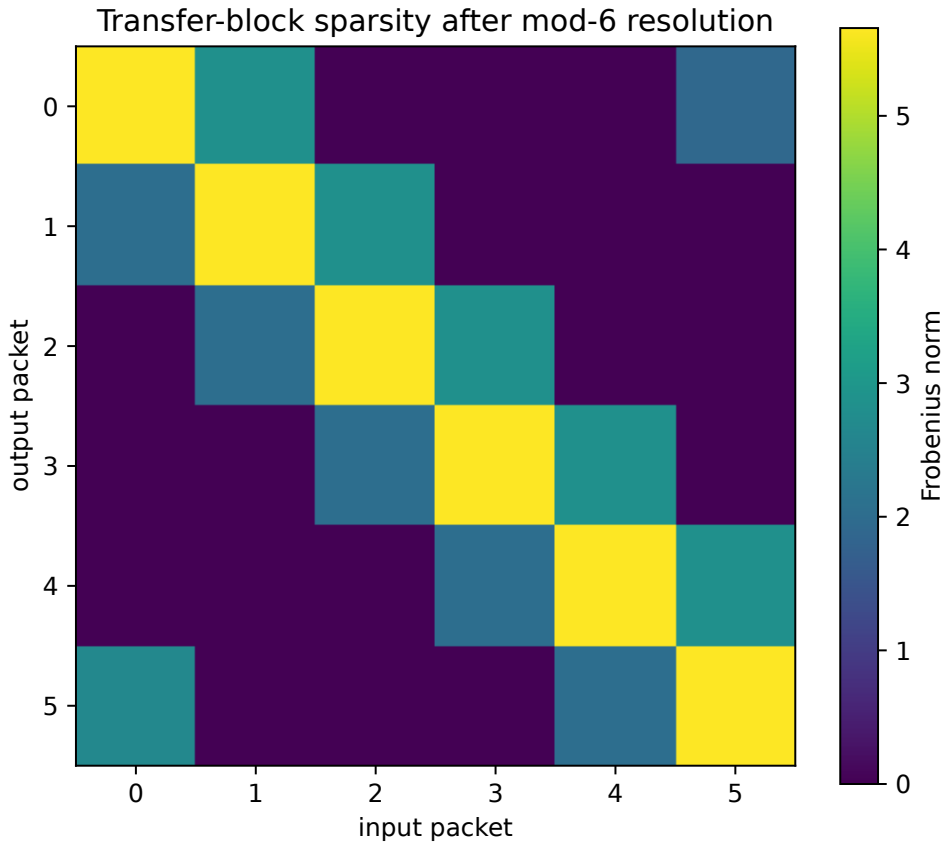


Figure 10: Packet-block norms for a finite stencil. The original operator is exactly reconstructed from a sparse transfer pattern.

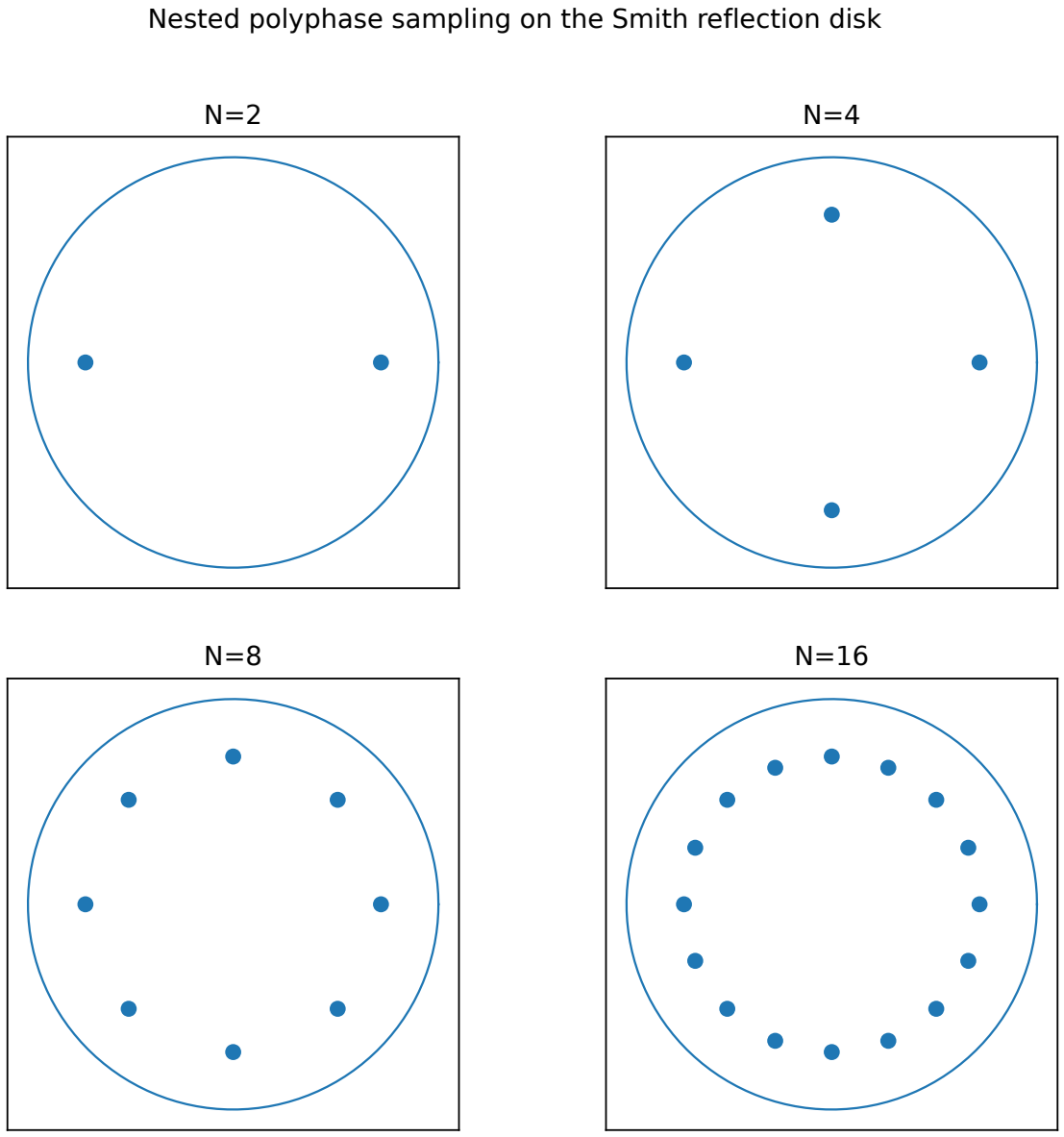


Figure 11: Nested 2-power phase orbits on the Smith reflection disk. The full Smith-chart treatment is deferred to the signal paper.

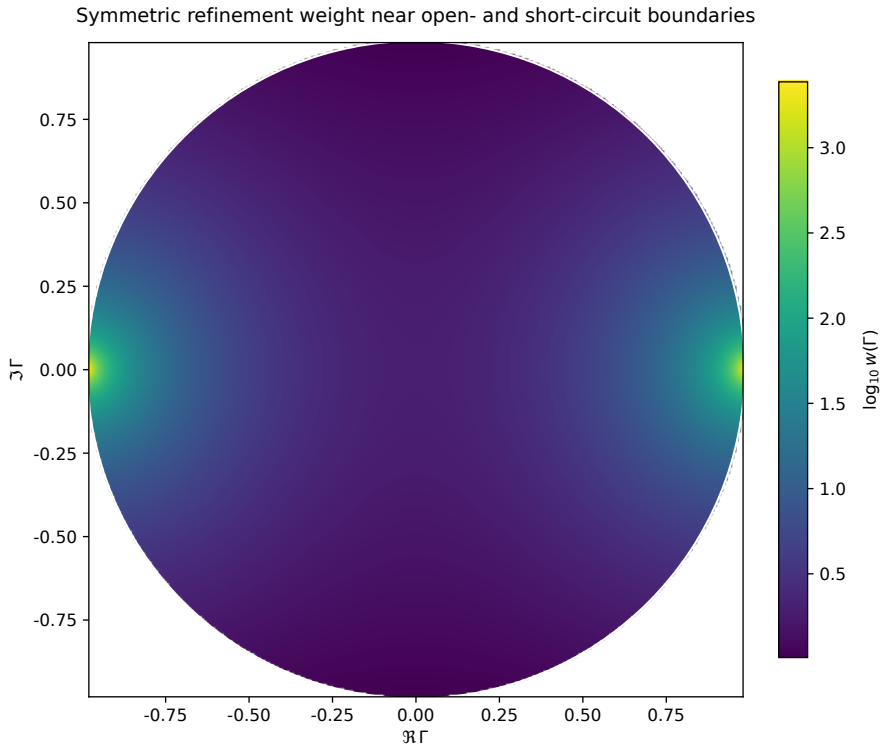


Figure 12: A symmetric refinement weight concentrated near the open- and short-circuit boundary points.

12 Numerical certificates

All numerical work in this paper is finite and reproducible. It tests the Python implementation of identities proved or certificate-checked above; it does not replace those arguments or the Lean kernel checks. Release 1.2.1 runs forty-eight tests, including invalid-input contracts, refinement transitivity, transfer-block aggregation, Fourier/DFT agreement, brute-force Smith fibre counts, and consistency of the stored JSON and CSV records.

Numerical Certificate 12.1 (Weighted inverse). For the tested diagonal-plus-transfer matrices, the filtered inverse agrees exactly with direct modular inversion at all tested precisions:

$$2^{10}, \qquad 3^{8}, \qquad 5^{7}.$$

The maximum modular residual is zero in every case.

Numerical Certificate 12.2 (Transfer zeta). The rational function

$$\frac{-t^3}{t^9 - t^7 - t^6 - t^5 + 2t^4 + t^3 + t^2 - 1}$$

was expanded through degree 30. Every coefficient agrees with direct weighted-walk enumeration.

Numerical Certificate 12.3 (Dyadic separation). The code verifies exactly that the two rational sign projectors are idempotent, orthogonal, and complementary, that scaling by 2 makes both integral, and that the same projectors are already integral modulo every tested odd prime.

Table 1: Representative numerical and symbolic checks.

Check	Result
Packet synthesis residuals for moduli $2, 3, 4, 6, 8, 12$	0
Nested refinement residuals for $2 \to 4$, $2 \to 8$, $3 \to 6$, $3 \to 12$, $4 \to 8$, $4 \to 12$, $6 \to 12$	0
Dyadic–odd CRT residuals for 4×3, 8×3, 4×5, 8×5	0
Maximum finite-Fourier projector residual	6.60×10^{-15}
Weighted inverse residual modulo 2^{10}	0
Weighted inverse residual modulo 3^8	0
Weighted inverse residual modulo 5^7	0
Difference between filtered and direct modular inverses	0
Transfer-zeta coefficients versus direct walk enumeration through weight 30	exact match
Dyadic sign-projector resultant weight	$v_2(2) = 1$
Odd-prime sign-projector resultant weight at 3	$v_3(2) = 0$
Power-defect telescoping identity residual	at most 6.94×10^{-18}
Resolvent-defect identity residual	1.09×10^{-17}
Smith $N = 2$ phase versus $z \mapsto 1/z$	residual below 1.4×10^{-16}

13 Boundaries and next questions

Boundary 13.1 (Refinement is not truncation). The exact refinement maps in this paper preserve the full object. A numerical scheme that discards fine packets requires an additional approximation theorem and cannot cite exact coordinate refinement as a truncation guarantee.

Boundary 13.2 (Linear versus nonlinear defects). The power and resolvent identities concern linear operators. For nonlinear maps, packet interactions require higher jets and operator trees. That theory is the subject of Paper III.

Boundary 13.3 (Resultants give bounds). A resultant supplies an integral denominator bound. It need not always be the minimal separation weight for a particular lattice, because cancellation can lower the actual denominator.

Boundary 13.4 (Dyadic arithmetic). The fact that a coefficient selector remains integral at 2 does not imply that every spectral, character, or deformation projector does. Each construction must be audited for division by the group order, resultants, trace splittings, and quadratic refinement data.

The next paper will add nonlinear operation trees, exact packet-coordinate composition, refinement defects for nonlinear maps, holonomy, and obstruction packets.

14 Conclusion

The central result of Paper II is that refinement can be made exact without erasing the distinctions that appear in integral arithmetic. The same underlying object can be represented by a nested tower of packet coordinate systems,

$$P_{\ell,\alpha} = \sum_{\beta \succ \alpha} P_{\ell+1,\beta},$$

and every linear operator refines by summing its fine transfer blocks. Exact coordinate changes obey identity and cocycle laws, while typed coordinate operators satisfy a naturality square. Residue selectors

are available over arbitrary coefficient rings; a finite-Fourier realization over a commutative semiring requires the stated finite orthogonality table, with the familiar primitive-root formula recovered over a field. A composite modulus factors into dyadic and odd packet coordinates, but their integral separation weights need not agree.

Certified weighted transfer words provide the finite-quotient inverse formula

$$q_L(T)^{-1} = q_L\left(\sum_{\gamma}(-1)^{|\gamma|}A_\gamma D^{-1}\right),$$

with only $W(\gamma) < L$ retained; a complete-normed contraction supplies the corresponding infinite Neumann inverse. Finite transfer graphs have an exact coefficient algorithm and a semantic first-edge recurrence. For positive integer weights, the determinant and adjugate of $I - A(t)$ automatically identify that enumerated series with a normalized rational expression and produce its eventual constant-coefficient recurrence. Minimal evaluated Bézout hypotheses give the separator numerator identity; explicit lattice-unit data turns it into an upward-closed separation bound. Certified Smith coordinates convert lifting weights into solvability thresholds and conditional branching counts, and concrete $\mathbb{Z}/p^L\mathbb{Z}$ fibres realize those certificates. The elementary involution example shows exactly why 2 must be treated separately: its sign projectors have exact local weight one at 2, equal to the valuation of their computed resultant, and weight zero at every odd prime.

The common operator language is therefore

packet → descendants → transfer words → weights → finite precision → exact reconstruction.

This language will be retained in every later realization.

The release-level contribution is therefore both mathematical and epistemic: each counted statement has a stable manuscript label, a precise Lean target, and a declared boundary separating constructed mathematics from supplied certificates and classical input. Kernel acceptance establishes internal formal correctness of those targets. It does not by itself establish novelty or replace independent mathematical review.

A Notation table

Symbol	Meaning
$P_{\ell,\alpha}$	packet projector at refinement level ℓ
I_ℓ	packet address set at level ℓ
$\beta \succ \alpha$	fine packet β is a child of coarse packet α
$\mathcal{A}_\ell, \mathcal{S}_\ell$	packet analysis and synthesis maps
$J_{\ell+1,\ell}$	exact coordinate refinement map
$C_{\ell,\ell+1}$	exact fine-to-coarse coordinate regrouping
$T^{(\ell)}_{\beta\alpha}$	transfer block $P_{\ell,\beta}TP_{\ell,\alpha}$
$P_{N,j}$	residue or phase selector for address $j \pmod N$
$w(e)$	positive filtration weight of a transfer edge
$W(\gamma)$	cumulative weight of a transfer word or graph walk
$A(t)$	polynomial weighted adjacency matrix
$Z_{s,t}(t)$	weighted transfer-zeta function from vertex s to vertex t
$s_{\pi,\Lambda}(E)$	lattice-dependent separation weight of a rational projector, when a clearing exponent exists

Symbol	Meaning
Δ	one-step linear refinement defect $CT_f - T_cC$

B Algorithmic forms

B.1 Residue analysis and synthesis

```
def analyze_residues(x, N):
    require N > 0
    return [x[j::N] for j in range(N)]

def synthesize_residues(packets):
    N = len(packets)
    require packet lengths fit one N-way interleaving
    y = empty(sum(len(p) for p in packets))
    for j, packet in enumerate(packets):
        y[j::N] = packet
    return y
```

The verified contract is synthesize(analyze$(x, N)) = x$. The Lean counterpart uses range-restricted packet coordinates and proves both inverse laws, while the finite Python routine checks the concrete array realization.

B.2 Weighted finite-precision inverse

```
Dinv = inverse(D)                       # certified on both sides
edge[e] = Dinv @ H_edge[e]
require Dinv @ H == sum(edge[e] for e in edges)
require edge[e] lies in filtration[weight[e]]
m = certified_lower_bound(weight)       # 1 <= m <= weight[e]
N = (L + m - 1) // m                    # so L <= m*N
S = 0
for n in range(N):
    for word in edge_tuples(edges, n):
        if sum(weight[e] for e in word) < L:
            S += (-1)**n * ordered_product(edge[e] for e in word)
return q_L(S @ Dinv)
```

Lean proves that the mapped pruned sum is exactly the mapped finite geometric inverse. The lower bound m gives $m|\gamma| \le W(\gamma)$, so $L \le mN$ derives $q_L(A^N) = 0$ from the filtration certificate and makes the returned value a two-sided inverse of $q_L(D + H)$. The Lean lemma $L \le m\lceil L/m \rceil$ justifies the displayed executable cutoff for $m > 0$. Taking $m = 1$ and $N = L$ gives the unconditional positive-weight cutoff. The Python implementation checks the DVR matrix specialization against direct modular inversion.

B.3 Transfer-zeta walk construction

```
def coefficient(G, start, target, n):
    total = 0
    for length in range(n + 1):
        for word in edge_tuples(G, length):
            if is_walk(word, start, target) and weight(word) == n:
                total += 1
```

```
    return total

A[i,j] = sum(t**weight(e) for distinct edges e from i to j)
T = I - A
Q = det(T)
H = adjugate(T)
P = H[start, target]
assert Q.coeff(0) == 1
Z = coefficient_series(G, start, target)
prove Q * Z == P
```

Positive weights make the coefficient loop finite because every contributing word has length at most n. Lean proves the enumeration correct, derives its first-edge recurrence, proves $(I - A)Z = I$, constructs the determinant/adjugate inverse, and identifies the two matrices by inverse uniqueness. Thus the normalized $QZ = P$ identity and its induced recurrence follow from finite graph data. The Python companion repeats the exact symbolic construction as an executable cross-check; it is not an oracle used by the Lean proof.

C Proof dependency map

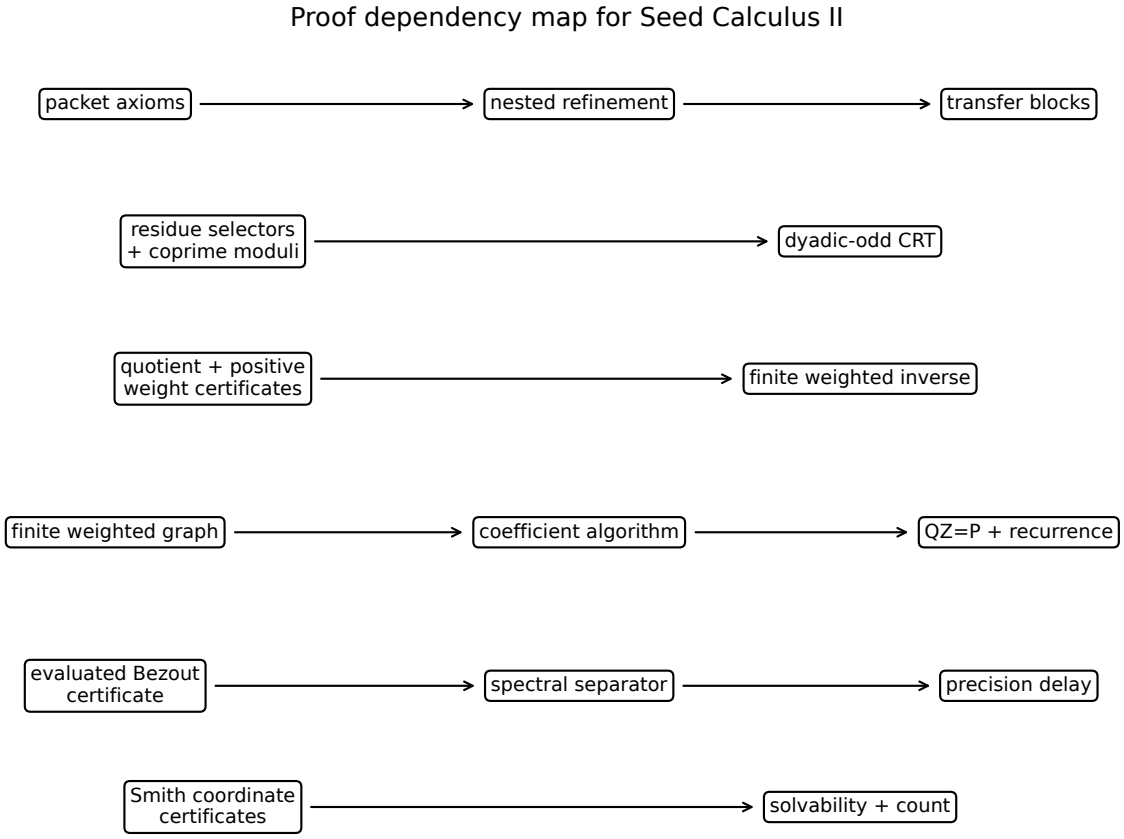


Figure 13: The dependency structure of Paper II. Coordinate coherence, residue/CRT, weighted inversion, automatic finite-graph transfer rationality, and integral separation are distinct branches with explicit inputs.

The proof order is:

complete packet axioms ⇒ finite-packet uniqueness and coordinate coherence
⇒ coordinate naturality and refined transfer blocks,
residue selectors and a finite Fourier table ⇒ Fourier formulas,
coprime moduli ⇒ dyadic–odd formulas,
quotient, filtration, and edge-weight certificates ⇒ explicit pruned inverse,
finite positive-weight graph ⇒ semantic enumeration and first-edge recurrence
⇒ det / adj rationality and recurrence,
minimal evaluated Bézout data and inverse-unit lattice action
⇒ separation bound, upward closure, and precision delay,
concrete $\mathbb{Z}/p^L\mathbb{Z}$ coordinate fibres ⇒ finite Smith product count,
one-step typed defect ⇒ powers, polynomial actions, and witnessed resolvents.

The transfer-zeta walk branch is logically independent of operator inversion. Its rationality conclusion now follows from finite graph data, rather than from a supplied $QZ = P$ premise. Likewise, CRT factorization follows directly from residue selectors and coprimality; it does not require the abstract refinement theorem, although it supplies a concrete refinement family. General Smith diagonalization and construction of polynomial Bézout/resultant data remain inputs upstream of the formalized post-certificate branches.

D Reproducibility layout

```
Projector_Resolved_Seed_Calculus_II_v1.2.1/
  paper/
    Projector_Resolved_Seed_Calculus_II.{tex,pdf}
    figures/                          # 13 PDF/PNG pairs
  formalization/lean/
    KomorebiVerification/PaperII/     # modules + KernelAudit
    lean-toolchain, lake-manifest.json
    LEAN_THEOREM_MAP_v1.1.1.md
  code/python/
    seed_refinement/, tests/, dist/
    pyproject.toml, requirements.{in,lock}
  results/                            # JSON, CSV, run logs
  provenance/                         # claim map and audits
  README.md, MANIFEST.md, RELEASE.json, CHANGELOG.md
  CITATION.cff, CITATION.bib, LICENSE_STATUS.md
  Run_All.{sh,ps1}, SHA256SUMS.txt
```

The Lean toolchain and complete dependency graph are pinned. The source audit rejects placeholders and project-specific axioms, the Python run uses an exact lock file, and the root checksum manifest covers every release file other than the manifest itself. The ongoing series checklist remains separate.

References

[1] Komorebi P., “Projector-Resolved Seed Calculus I: Typed Seeds, Operator Grammars, and Exact Reconstruction,” version 1.2.1, manuscript and verification release, 2026.

[2] L. de Moura, S. Kong, J. Avigad, F. van Doorn, and C. Roux, "The Lean theorem prover (system description)," in *Automated Deduction—CADE-25*, Lecture Notes in Computer Science 9195, pp. 378–388, Springer, 2015. https://doi.org/10.1007/978-3-319-21401-6_26.

[3] The Mathlib Community, "The Lean mathematical library," in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pp. 367–381, 2020. https://doi.org/10.1145/3372885.3373824.

[4] J.-P. Serre, *Local Fields*, Graduate Texts in Mathematics 67, Springer, 1979. https://doi.org/10.1007/978-1-4757-5673-9.

[5] S. Lang, *Algebra*, 3rd ed., Graduate Texts in Mathematics 211, Springer, 2002. https://doi.org/10.1007/978-1-4613-0041-0.

[6] R. P. Stanley, *Enumerative Combinatorics*, vol. 1, 2nd ed., Cambridge University Press, 2012. https://doi.org/10.1017/CBO9781139058520.

[7] P. P. Vaidyanathan, *Multirate Systems and Filter Banks*, Prentice Hall, 1993. ISBN 978-0-13-605718-5.

[8] D. M. Pozar, *Microwave Engineering*, 4th ed., Wiley, 2012. ISBN 978-0-470-63155-3.